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TRIGEMINAL NEURALGIA AND ITS MANAGEMENT

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Nov, 2025

OUTLINE

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- ✓ Anatomy
- ✓ Definition
- ✓ Etiology and Pathophysiology
- ✓ General Characteristics
- ✓ Clinical Characteristics
- ✓ Diagnosis
- ✓ Treatments

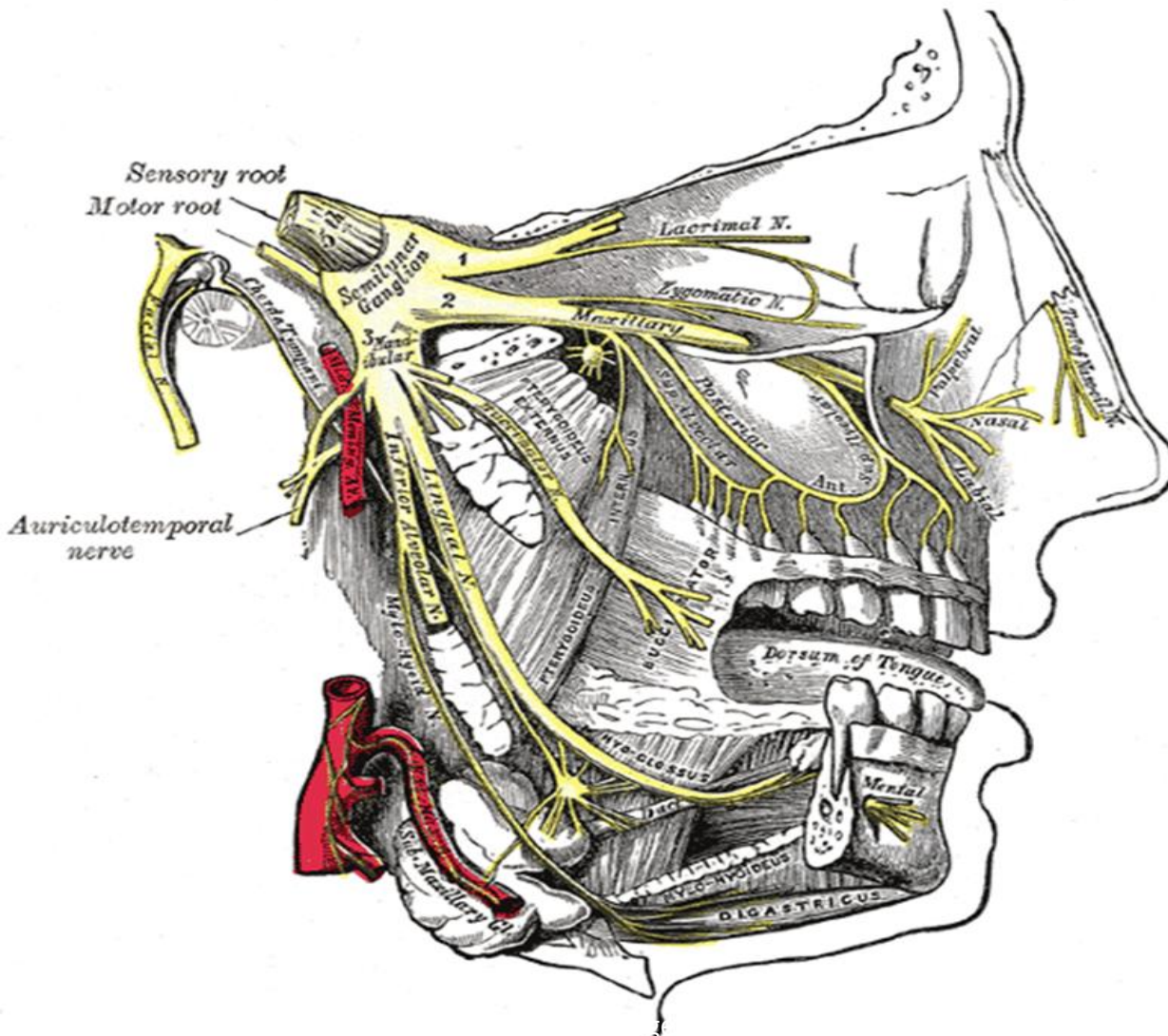
Discussion

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- Anatomy of the trigeminal nerve?

Trigeminal Nerve Anatomy

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Definition

- Trigeminal neuralgia (TN) is defined as sudden, usually unilateral, severe, brief, stabbing, lancinating, paroxysmal, recurring pain in the distribution of one or more branches of 5th cranial nerve.

HISTORY

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- John Locke in 1677, gave the first full description with its treatment.
 - Nicolaus Andre in 1756, coined the term ‘Tic Doulo ureux.’
 - John Fothergill in 1773, published detailed description of TN, since then, it has been referred to as ‘Fothergill’s disease’.
- Is there any other name?

Etiology And Pathophysiology

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- TN is caused by a proximal compression of the trigeminal nerve root close to the brainstem (root entry zone) by a tortuous or ectasic blood vessel (an artery or vein), leading to mechanically twisted nerve fibers and secondary demyelination, probably mediated by microvascular ischemic damages.

Cont.

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- These changes lower the excitability threshold of affected fibers and promote inappropriate ephaptic propagation toward adjacent fibers.
- Thus, tactile signals coming from the fast myelinated (A-beta) fibers can directly activate the slow nociceptive (A-delta) fibers, resulting in the high-frequency discharges characteristic of trigeminal neuralgia.

Cont.

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- One theory suggests that peripheral injury or disease of the trigeminal nerve increases afferent firing in the nerve, perhaps by ephaptic transmission between afferent unmyelinated axons and partially damaged myelinated axons;
- Failure of central inhibitory mechanisms may also be involved.
- The attacks of pain in TN are probably caused by a group of pathologically “hyperexcitable” sensory trigeminal neurons that discharge spontaneously, exhibit abnormal after discharges, and demonstrate abnormal spread of excitation to adjacent primary afferents.

Etiologies

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I. Dental etiology

Westrum and Black (1976), differentiation from loss of teeth and degeneration of nerve is not restricted to peripheral parts of the ganglia, but proceeds proximally to involve areas of spinal nucleus.

II. Infections: Various granulomatous and non - granulomatous infections.

Cont.

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III. Multiple sclerosis: Olfson (1966), suggested the presence of sclerotic plaque located at the root entry zone of the trigeminal nerve.

IV. Petrous ridge (basilar) compression

V. Post traumatic neuralgia

Cont.

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VI. Intracranial tumors: Many lesions such as epidermoid tumors, meningiomas of cerebellopontine angle and Meckel's cave,

VII. Intracranial vascular abnormalities
arteriovenous malformations

aneurysms and vascular compression

VIII. Viral etiology

General Characteristics

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- Incidence: 4 in 100,000 persons.
- Age : Late middle age or later in life (5th or 6th decade)
- Sex : With female predisposition (58%).
- Affliction for sides: Predilection for the right side is noted(60%)
- Division of Tn: $V_3 > V_2$ Very rarely $V_1 < 5\%$

Clinical Characteristic

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- ✓ Pain is usually confined to one part of one division of trigeminal nerve.
- ✓ Pain rarely crosses the midline.
- ✓ The pain is of short duration and lasts for a few seconds, but may recur with variable frequency.
- ✓ Patient grimaces with pain, clutches his hands over the affected side of the face.

Cont.

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- ✓ Typically arises in the persons who have no abnormal neurologic deficit.
- ✓ The paroxysms occur in cycles, each cycle lasting for weeks or months and with time, the cycle appears closer and closer.
- ✓ In extreme cases, the patient will have a motionless face—the ‘frozen or mask like face’.

Cont.

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- ✓ Presence of an intraoral or extraoral trigger points provokable by obvious stimuli is seen in TN.
- ✓ The location of the trigger points
 - ❖ In V2—points are located on the skin of the upper lip, ala nasi or cheek or on the upper gums.
 - ❖ In V3—this is the most frequently involved branch.
 - Trigger points are seen over the lower lip, teeth or gums of the lower jaw.
 - Tongue is rarely involved.
 - ❖ In V1—the trigger zone usually lies over the supraorbital ridge of the affected side.

Cont.

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- ✓ It is characteristic of the disorder, that attacks do not occur during sleep.
- ✓ Poor quality of life.
- ✓ Undergo indiscriminate dental extractions on the affected side without any relief from pain.
- ✓ More than 50% of patients experience early remissions of greater than 6 months before return of active pain.

Diagnosis

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- The diagnosis of TN rests on the clinician`s ability in recognizing a distinctive series of signs and symptoms that define this disorder.
- White and Sweet made a significant contribution by articulating precise diagnostic criteria for TN.
- Consists of 5 major clinical features that define the diagnosis of TN
- The Sweet criteria were incorporated into the criteria published by IASP & IHS.

Sweet diagnostic criteria

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1. Pain is paroxysmal
2. The pain may be provoked by light touch to the face (trigger zones)
3. The pain is confined to the trigeminal distribution
4. The pain is unilateral.
5. The clinical sensory examination is normal

Cont....

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- The current IHS recommended classification for TN is listed in “International Classification of Headache Disorders II ICHDII”.
- It subdivides TN into “Classic idiopathic TN” and “Symptomatic TN” based on presumed etiologies.
- Classic TN encompasses both idiopathic TN cases and those related to vascular compression.
- Secondary (or symptomatic) TN is reserved for patients with TN caused by structural lesions other than vascular compression.

Classical Criteria

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- A. Paroxysmal attacks of pain lasting from a fraction of a second to 2 minutes, affecting 1 or more divisions of the trigeminal nerve, & fulfilling criteria B & C.
- B. Pain has at least 1 of the following characteristics:
 - 1. Intense, sharp, superficial, or stabbing
 - 2. Precipitated from trigger zones or by trigger factors

Cont....

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- C. Attacks are stereotyped in the individual patient.
- D. No clinically evident neurologic deficit.
- E. Not attributed to another disorder.

Symptomatic Criteria

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- A. Paroxysmal attacks of pain lasting from a fraction of a second to 2 minutes, **with or w/o persistence of pain between paroxysms**, affecting 1 or more divisions of the trigeminal nerve, & fulfilling criteria B & C.
- B. Pain has at least 1 of the following characteristics:
 - 1. Intense, sharp, superficial, or stabbing
 - 2. Precipitated from trigger zones or by trigger factors

Cont....

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C. Attacks are stereotyped in the individual patient

D. **A causative lesion**, other than vascular compression, **has been demonstrated** by special investigations &/or posterior fossa exploration.

- Proper clinical examination along with well taken history is mandatory.
- HEENT, including TMJ and Masseter
- Oral exam, especially teeth and gum.
- Nuero exam
- Check for trigger zones
- Searching for red flags

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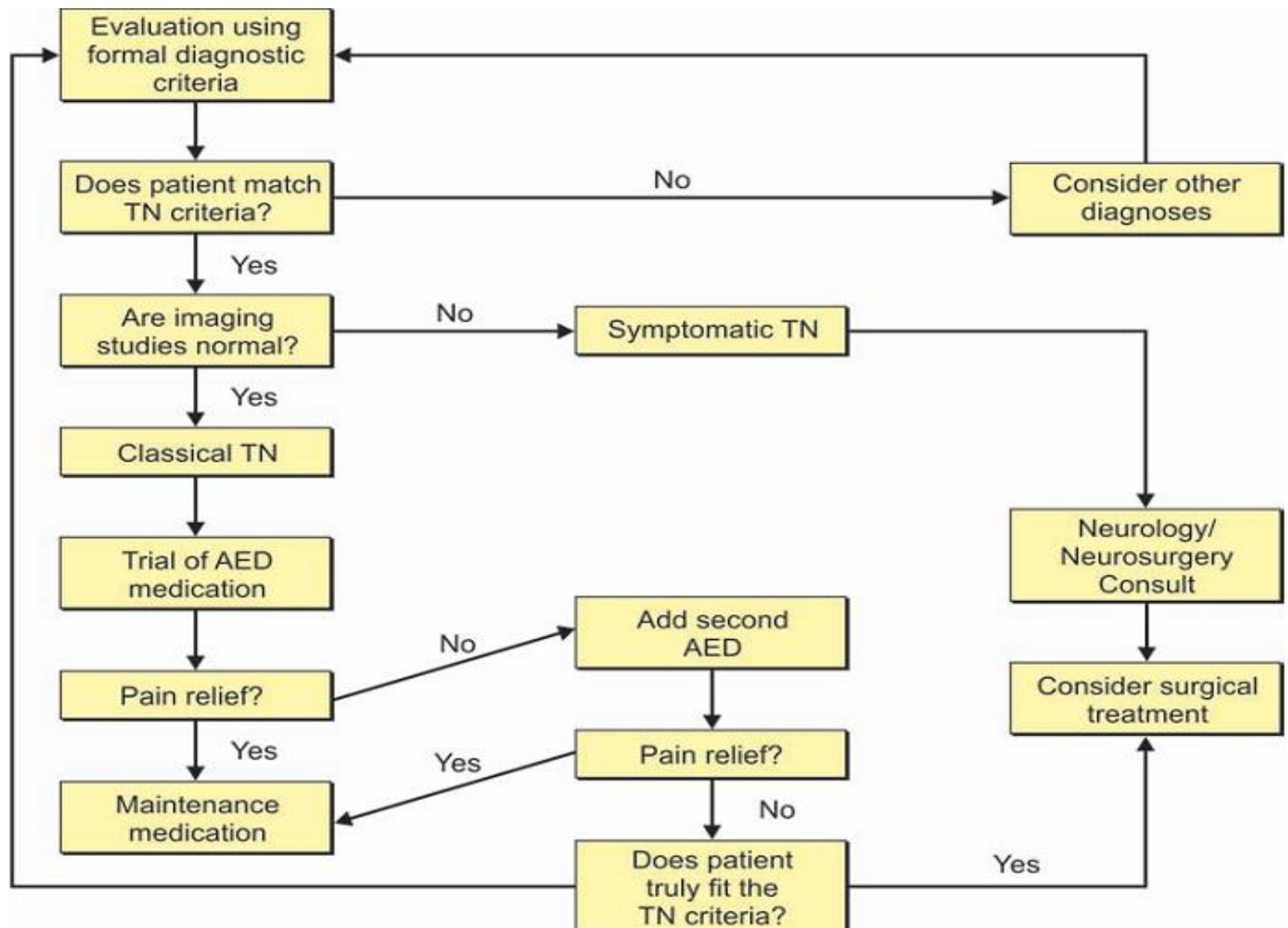
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- The neuralgic symptoms in younger group of patients (<35years of age) should alert the clinician to a possible intracranial space occupying lesion or intracranial arteriovenous anomalies.
- Other DDX acoustic neurilemoma, multiple sclerosis, post herpetic neuroma or post traumatic neuralgias

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- All patients should ideally have MRI scanning or at least a CT scan.
- Response to treatment with tablet carbamazepine is universal in trigeminal neuralgia diagnosis.
- Mathews and Scrivani have proposed a simple algorithm for the differential diagnosis and management of TN.



Rx -Medical

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- For patients with classic TN who require pain control, [carbamazepine](#) is initial first line therapy.
- For patients with classic TN who require pain control who do not respond to or tolerate [carbamazepine](#), switch to [oxcarbazepine](#).
- For patients who are refractory to or intolerant of [carbamazepine](#) and [oxcarbazepine](#), switching to treatment with [baclofen](#) .
- Alternatively, [lamotrigine](#) can be used as add-on therapy.

Pharmacological Treatment

- **AEDs**

- Tegretol (carbamazepine)
- Tripeptal (oxcarbazepine)
- Dilantin (phenytoin)
- Neurontin (gabapentin)
- Lyrica (pregabalin)
- Lamictal (lamotrigene)
- Topamax (topirimate)
- Gabatril (tiagabine)
- Keppra (levateracitam)

- **TCA's**

- Elavil (amitriptyline)
- Pamelor (nortriptyline)
- Desipramine (norpramin)

- **Baclofen (lioresal)**

Adverse Effects of AEDs

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- Cognitive changes
- Sedation
- Nystagmus, ataxia, diplopia, dizziness
- Nausea, vomiting, headache
- Allergic reaction
 - Up to 7% with CBZ

Surgical

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- Indicated when pharmacotherapy is ineffective or when its side effects markedly impair the patient's quality of life.
- The type of surgery depends on the patients medical condition, age and the clinical set available.

Micro vascular decompression

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- Micro vascular decompression provides the most sustained pain relief, with 90% of patients reporting initial pain relief and over 80% remaining pain free after 1 year, 75% after 3 years, and 73% after 5 years.
- It is, however, a major surgical procedure that entails craniotomy to reach the trigeminal nerve in the posterior fossa.
- Directly treats the hypothetical cause while minimizing any sensory damage.
- Is performed under GA through an opening in the posterior cranial fossa via suboccipital craniotomy and is widely accepted as a curative treatment method

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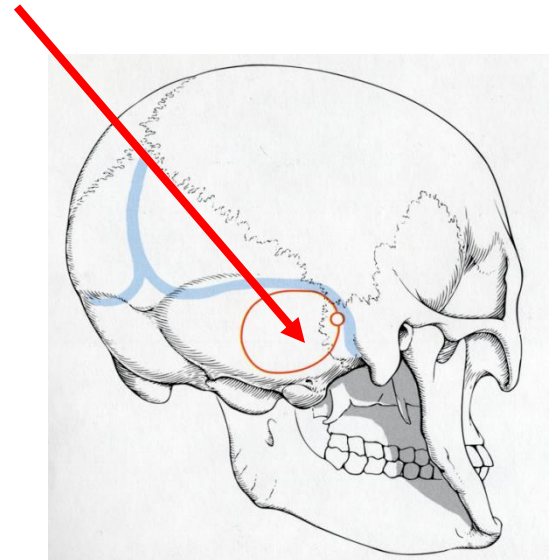
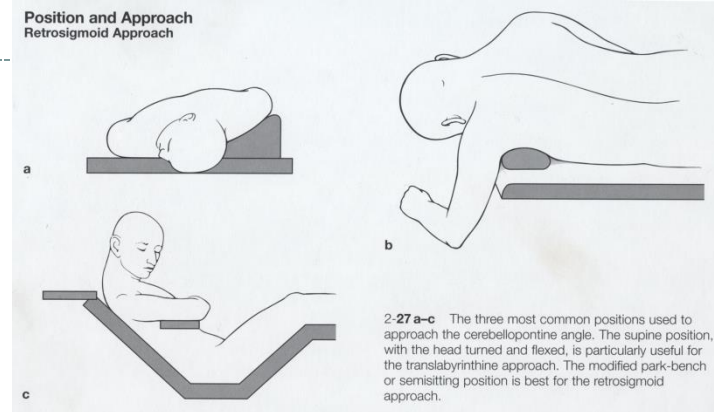
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- A contact between the trigeminal nerve and a blood vessel, usually the superior cerebellar artery, is found at the site where the nerve enters the brainstem .
- The two contacting structures are separated by the insertion of a piece of synthetic material, e.g., a small Teflon sponge.
- The average mortality rate ranges from 0.2% to 0.5%, and up to 4% of patients suffer from major problems such as cerebrospinal fluid leakage, infarcts, or hematomas.
- The most common long-term complications include aseptic meningitis (11%), sensory loss (7%), and hearing loss (10%).

Surgical Technique

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- Positioning
- Skin Incision
 - Inniomeatal line – transverse sinus
 - Digastric groove
 - $\frac{3}{4}$ - $\frac{1}{4}$ rule
- Retromastoid craniectomy
 - Expose sigmoid-transverse sinus junction
 - Mastoid emissary vein
 - Bevel bone laterally
 - Sufficient anterior exposure reduces amount of cerebellar retraction
- T-shaped dural opening
 - Exposure of most superior and lateral corner



Surgical Technique

- Exposure of CPA

- *“Turning the corner is the most dangerous stage of the operation and must be executed with patience and the utmost care” (Peter Jannetta)*

- CSF drainage
- Gentle retraction of ala of cerebellum
- Identify and divide petrosal vein

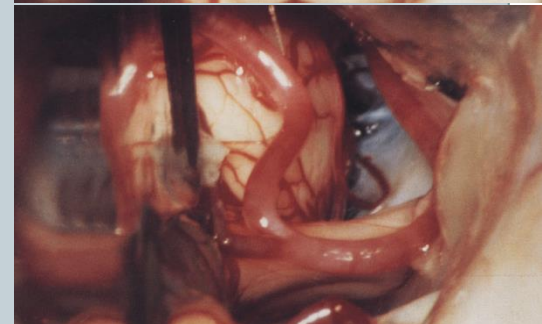
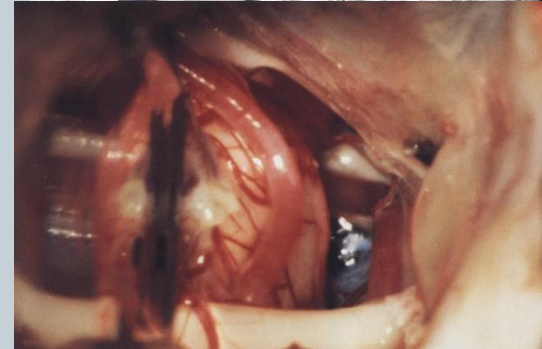
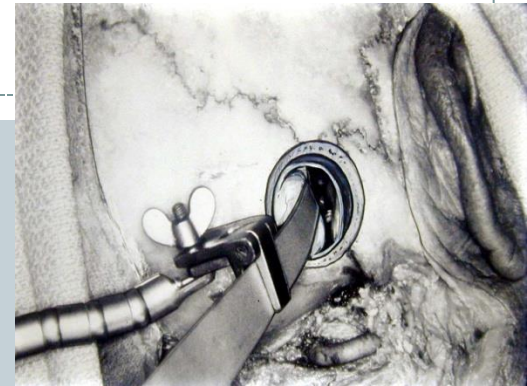
- Visualization of trigeminal nerve

- *Visualize the ENTIRE nerve from it's exit from the pons to it's exit laterally from the CPA*

- Decompression

- Mobilize and “pad” arteries
- Coagulate and divide veins

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percutaneous techniques

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- Gasserian ganglion percutaneous techniques are destructive interventions that include radiofrequency thermocoagulation, balloon compression, and percutaneous glycerol rhizolysis.
- 90% of patients report pain relief following these procedures.
- One year following radiofrequency thermocoagulation, 68–85% of patients are still pain free, but after 3 years the percentage goes down to 54–64%, and after 5 years only 50% of patients are still pain free.

Cont....

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- .The most common side effects are sensory loss (50%), dysesthesias (6%), anesthesia dolorosa (4%), and corneal numbness with a risk of keratitis (4%).
- Gasserian ganglion therapies require short-acting anesthetics and are primarily minor overnight procedures with an extremely low mortality rate.

Percutaneous techniques

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- A lesion is made in the trigeminal ganglion either with heat (thermocoagulation) or through a chemical effect (glycerol rhizolysis).
- Local anesthesia or transient intravenous sedation is given, and a special type of needle is introduced freehand, under fluoroscopic guidance, at a cutaneous puncture site 2 to 3 cm from the corner of the mouth, through the cheek, and into the foramen ovale.

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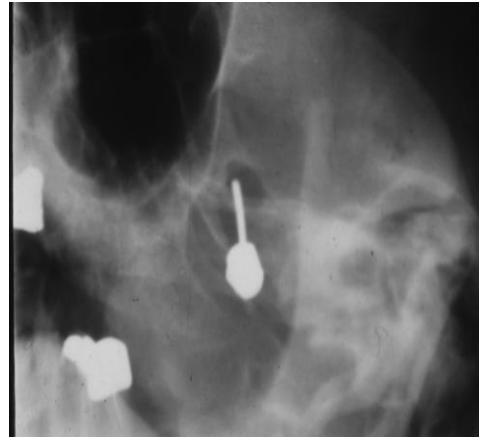
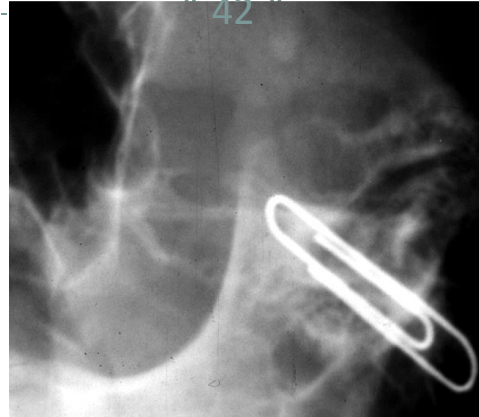
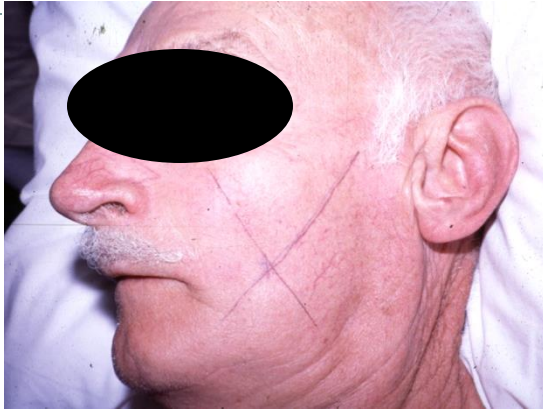
- The trigeminal ganglion is heated to 60° or 70°C for 60 to 70 seconds with a radiofrequency probe, producing a partial lesion, or else 0.3 to 0.4 mL of pure anhydrous glycerol is injected into Meckel's cave through the needle.

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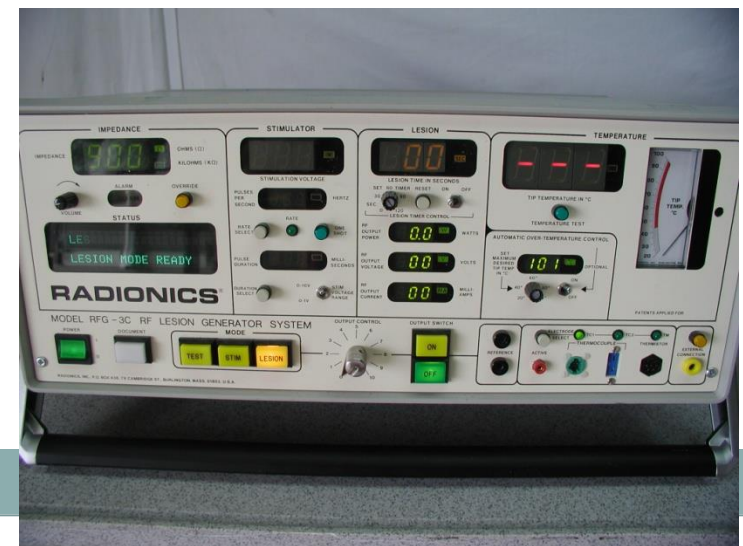
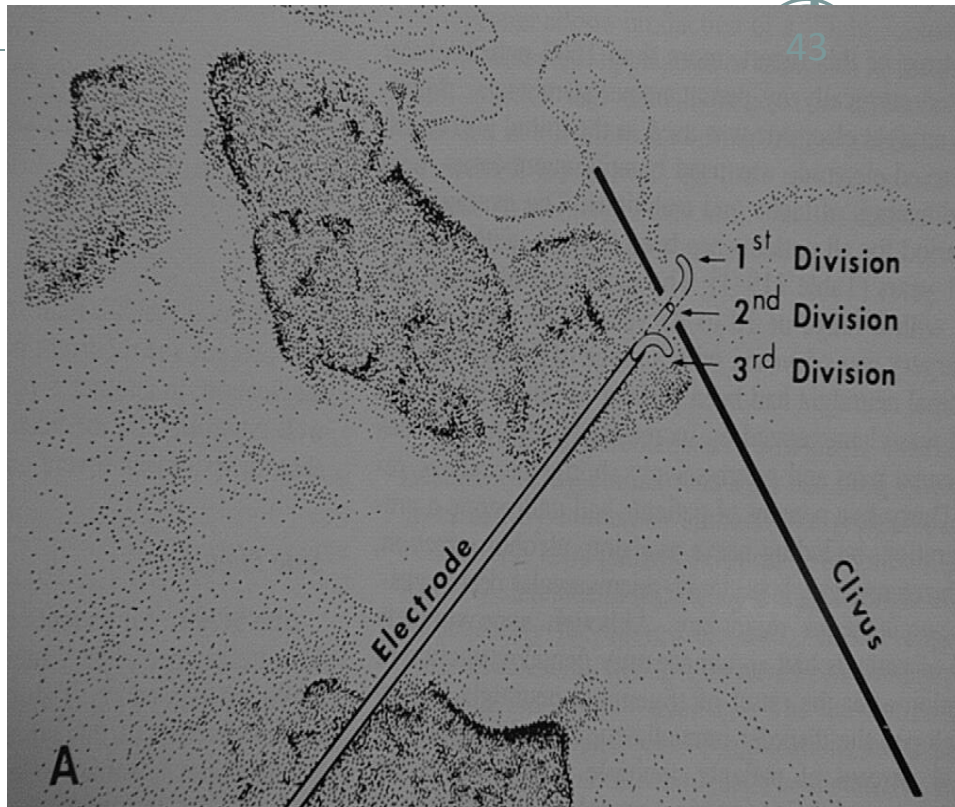
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- Balloon compression of the trigeminal ganglion is a third percutaneous technique that once enjoyed a certain degree of popularity.
- In this method, a 4 French Fogarty catheter is inserted through the foramen ovale and inflated for a few minutes with 0.75 to 1 mL of contrast medium (producing an intraluminal pressure of ca. 1500 mm Hg) .
- Hardly any long-term follow-up studies of this technique are available.

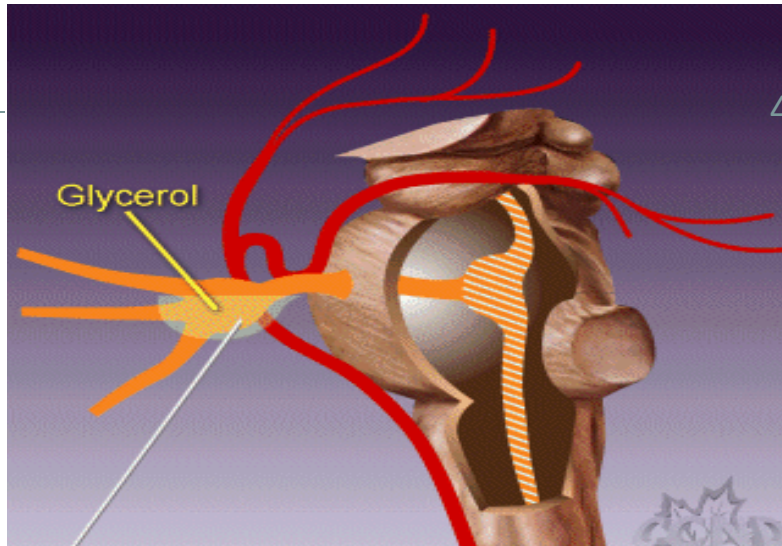
Needle Insertion



Radiofrequency Lesion



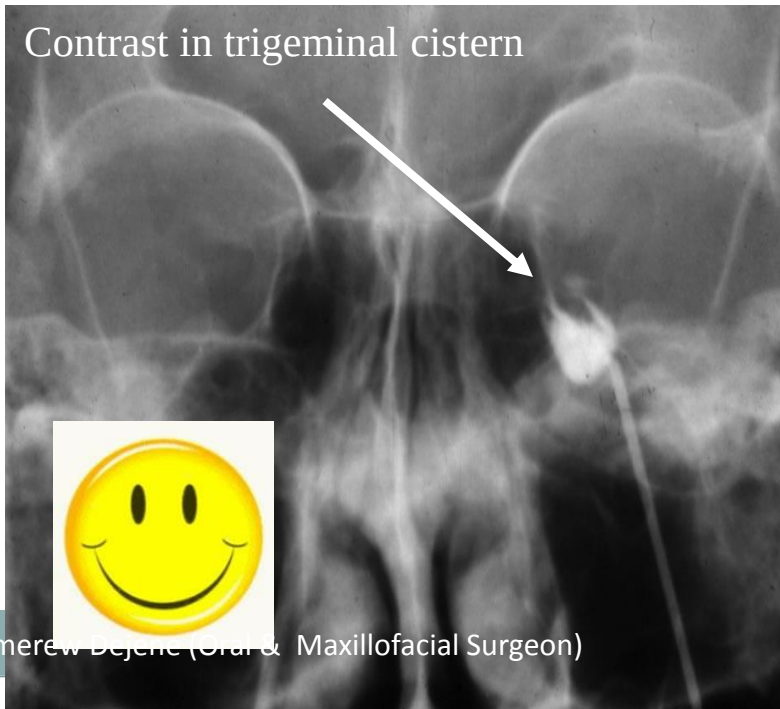
Glycerol Injection



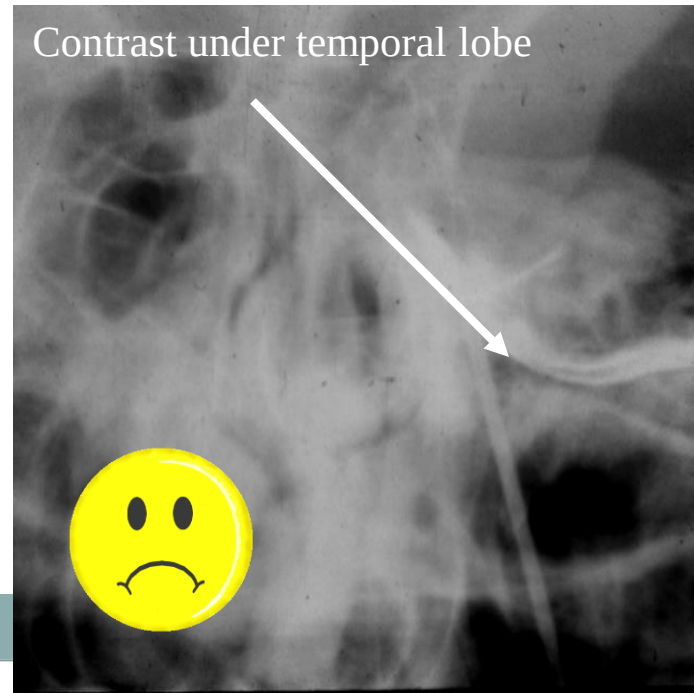
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Contrast in trigeminal cistern

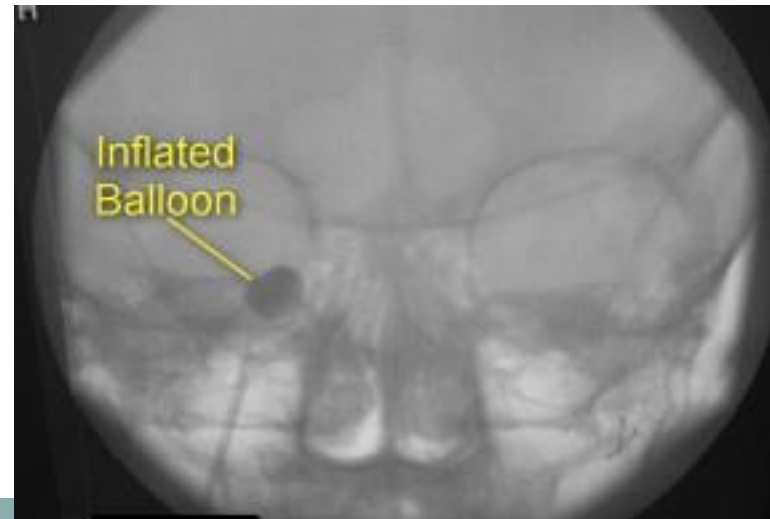
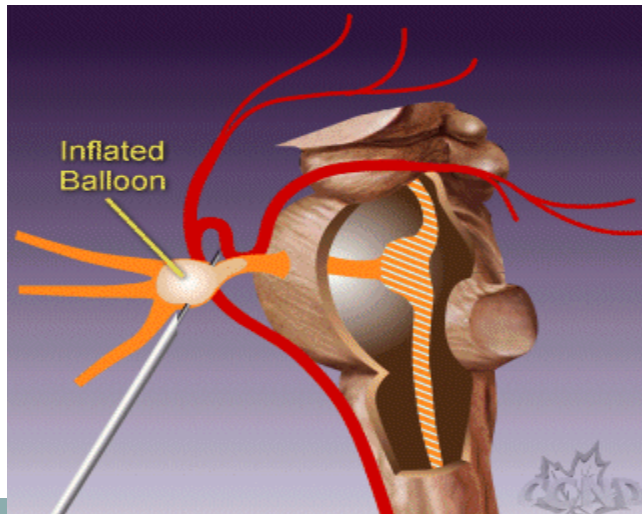
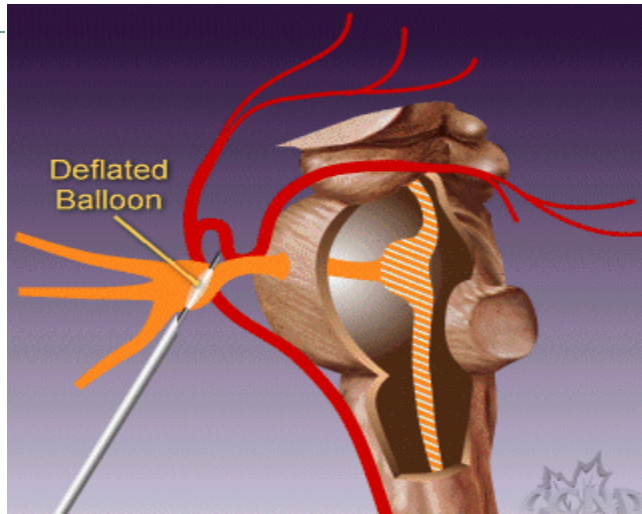


Contrast under temporal lobe



Balloon Compression

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MVD vs. Percutaneous Procedures

INITIAL PAIN RELIEF

- MVD 98%
- RF rhizotomy 98%
- Balloon 93%
- Glycerol 91%

RECURRENCE RATES

- Glycerol 54% (4 years)
- RF rhizotomy 23% (9 years)
- Radiosurgery 25% (3 years)
- Balloon 21% (2 years)
- MVD 15% (5 years)

Gamma knife surgery

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- Relatively recent
- Employs computerized stereotactic methods to concentrate a beam of gamma radiation aimed at the trigeminal root entry zone in the posterior fossa.
- One year after gamma knife surgery, 69% of patients were pain free without additional medication.
- After 3 years, 52% were still pain free.
- The development of pain relief can be delayed (for an average of 1 month).

Cont....

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- Side effects include sensory complications in 6% that may develop with a delay of up to 6 months, facial numbness in 9–37%, which improved over time, and paresthesias in 6–13%.
- Quality of life improved by 88%.
- The main disadvantage of gamma knife surgery is the cost, which limits its widespread use and makes it a last-reserve option for patients who cannot undergo open surgery or who have blood coagulation problems (e.g., patients who are taking warfarin).

Peripheral neurectomy

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- The most oldest and effective procedures for patients unresponsive to medication especially elderly and debilitated patients who can't tolerate posterior fossa exploration.

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